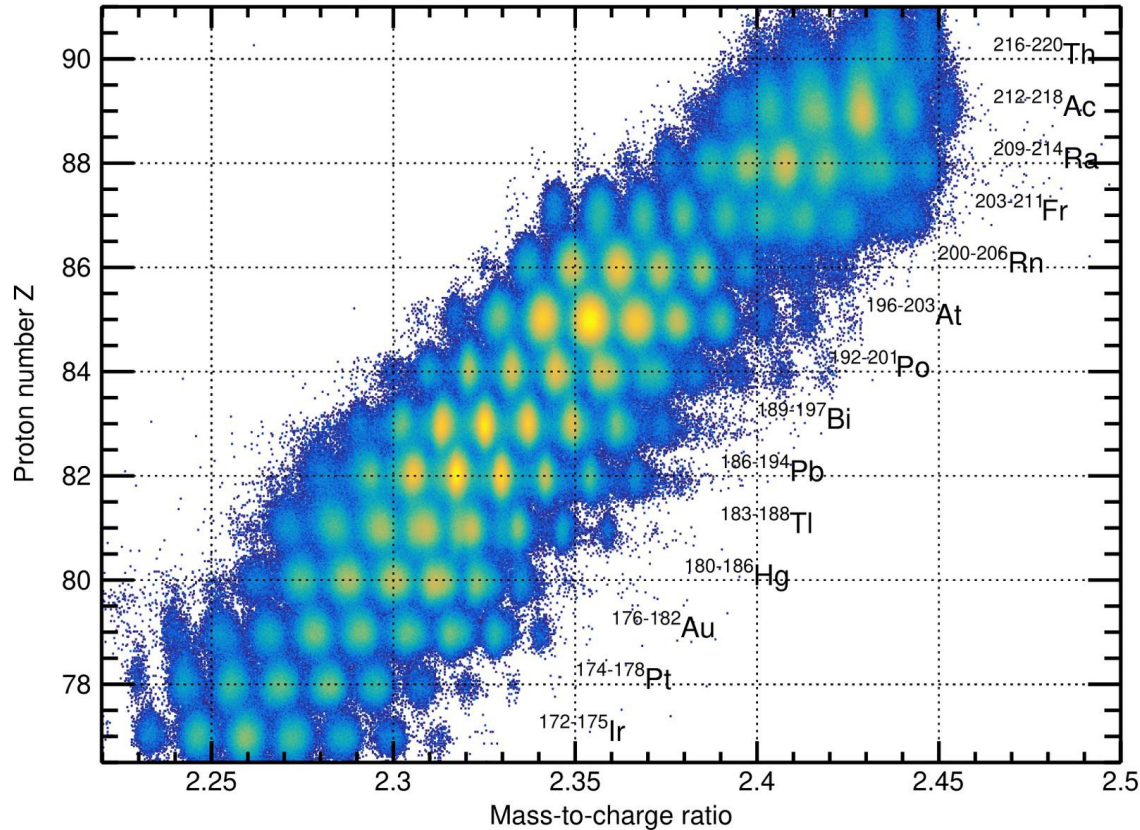


Fission investigated at FAIR Phase-0 using the advanced SOFIA@R3B setup



S455 experiment

- Fission study of 100 neutron-deficient exotic nuclei from iridium to thorium (12 FRS settings).
- Fission induced by Coulomb excitation.
- Detection of both fission fragment in coincidence.

Motivation

- Unexpected asymmetric fission in ^{180}Hg .
→ What are the limits of the new asymmetric fission island ?
→ What shell effect are responsible for the asymmetric split

Article

An asymmetric fission island driven by shell effects in light fragments

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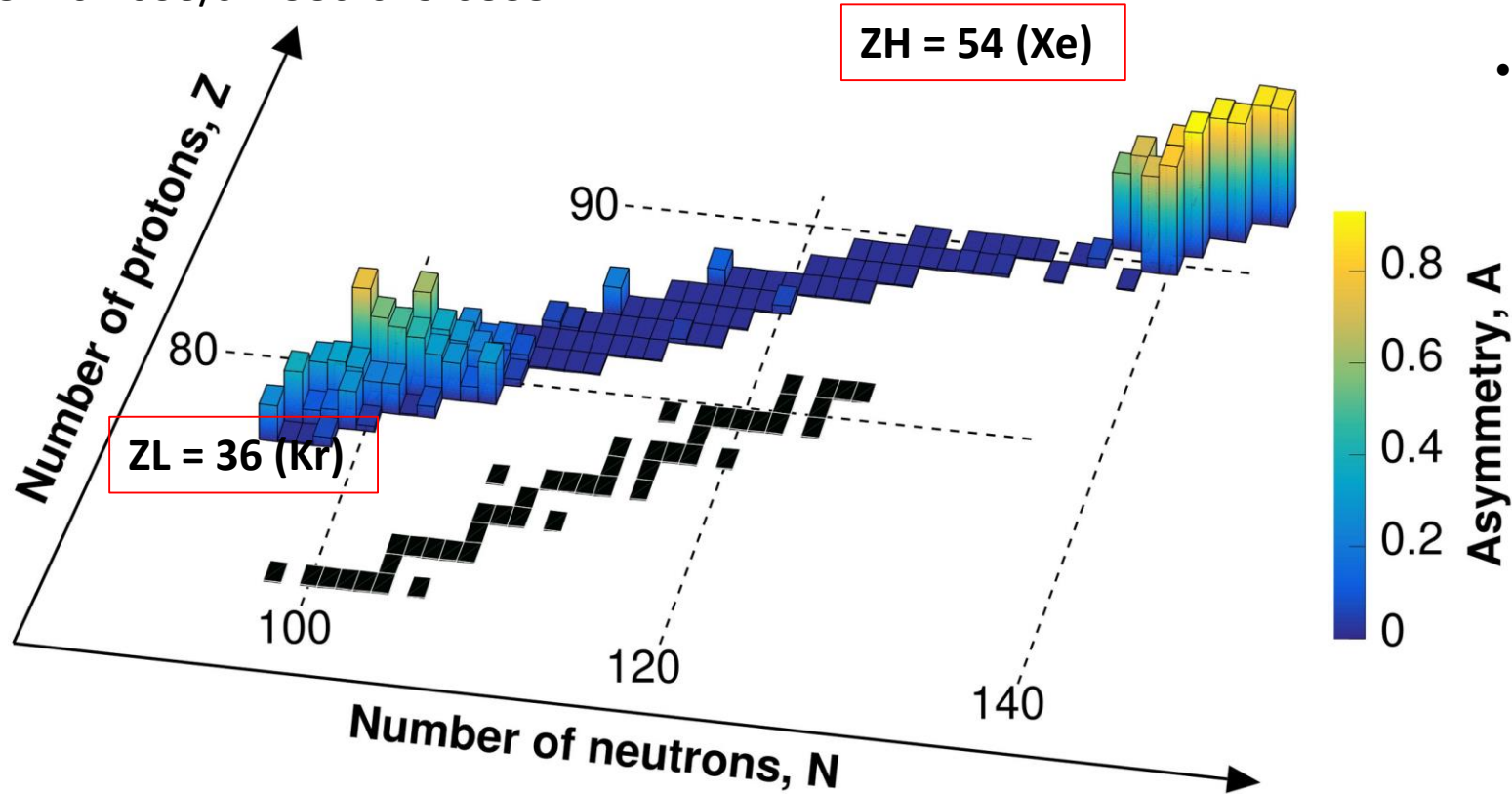
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A New Island of Asymmetric Fission

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Unlike the actinides, the asymmetric fission in the neutron deficient sub-lead region is driven by shell effect in the light fragment.

Results

- First set of charge distribution across a wide range of exotic fissioning systems.
- Experimental evidence of strong proton-shell stabilization at $Z=36$

